Physics: Work

Course Information

Course Name: Physics – Work

Level: Beginner / Intermediate

Prerequisites: Basic mathematics and an understanding of force and motion

Estimated Duration: 6–8 Hours

Course Description

Work is a fundamental concept in physics that describes the transfer of energy when a force causes an object to move through a displacement. In this course, students will learn the physical meaning of work, the relationship between force and displacement, positive and negative work, work done by constant and variable forces, work against gravity, and the relationship between work and energy. Students will also solve numerical problems involving work and explore real-world applications.

Learning Outcomes

After completing this course, students will be able to:

- Define work in physics.
- Explain the relationship between force and displacement.
- Identify the conditions required for work to be done.
- Calculate work done by a constant force.
- Understand the SI unit of work.
- Distinguish between positive, negative, and zero work.
- Understand the effect of the angle between force and displacement.
- Calculate work when force acts at an angle.
- Understand work done against gravity.
- Relate work to energy.
- Apply the work-energy principle.
- Solve numerical problems involving work.
- Identify examples of work in everyday situations.

Lesson 1: Introduction to Work

Learning Objectives

After this lesson, students will be able to:

- Define work in physics.
- Understand why force and displacement are both important.
- Distinguish between everyday and scientific meanings of work.

What is Work?

In everyday language, work can mean any physical or mental activity.

In physics, however, **work has a specific meaning**.

Work is done when a force acts on an object and causes the object to move through a displacement.

For example, if a person pushes a box and the box moves forward, the person does work on the box.

Conditions for Work

For mechanical work to occur:

1. A force must act on the object.
2. The object must undergo displacement.
3. The force must have a component in the direction of displacement.

If there is no displacement, mechanical work is zero.

Lesson 2: Work and Force

Work depends on both force and displacement.

For a constant force acting in the same direction as the displacement:

Work = Force × Displacement

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The standard formula is:

```
W = Fd
```

where:

- `W` = work
- `F` = force
- `d` = displacement

Example

A person pushes a box with a force of:

```
20 N
```

and the box moves:

```
5 m
```

Therefore:

```
W = Fd
```

```
W = 20 × 5
```

```
W = 100 J
```

The work done is:

```
100 joules
```

Lesson 3: SI Unit of Work

The SI unit of work is the **joule (J)**.

One joule is the work done when a force of one newton moves an object one meter in the direction of the force.

```
1 J = 1 N·m
```

Therefore:

```
Work = Force × Displacement
```

has units:

```
N × m = J
```

Lesson 4: Work as Energy Transfer

Work is closely related to energy.

When work is done on an object, energy can be transferred to or from that object.

For example:

- Pushing a box transfers energy to the box.
- Lifting an object transfers energy into gravitational potential energy.
- Friction can transfer mechanical energy into thermal energy.

Therefore, work can be understood as a **way of transferring energy**.

Lesson 5: Positive Work

Work is **positive** when the force and displacement have components in the same direction.

Example:

A person pushes a box forward, and the box moves forward.

The applied force and displacement point in the same direction.

Therefore:

```
Work > 0
```

Positive work generally increases the object's kinetic energy when it is the relevant net work.

Example

A force of:

```
50 N
```

moves an object:

```
4 m
```

in the direction of the force.

```
W = Fd
```

```
W = 50 × 4
```

```
W = 200 J
```

Lesson 6: Negative Work

Work is **negative** when the force has a component opposite to the displacement.

A common example is friction.

Suppose a box moves forward while friction acts backward.

The displacement is forward, while friction acts backward.

Therefore, friction does negative work.

```
W < 0
```

Negative work removes mechanical energy from the object's motion.

Example

A friction force of:

```
10 N
```

acts opposite to the displacement of:

```
6 m
```

Therefore:

```
W = -Fd
```

```
W = -(10)(6)
```

```
W = -60 J
```

Lesson 7: Zero Work

Work is zero when:

- There is no displacement.
- There is no force.
- The force is perpendicular to the displacement.

Example 1: No Displacement

A person pushes against a wall, but the wall does not move.

There is force, but:

```
Displacement = 0
```

Therefore:

```
W = 0
```

Example 2: Perpendicular Force

Suppose an object moves horizontally while a force acts vertically.

The force is perpendicular to the displacement.

Therefore, that force does zero work.

Lesson 8: Work Done at an Angle

Sometimes force does not act in exactly the same direction as displacement.

If the angle between force and displacement is `θ`, the work is:

```
W = Fd cos(θ)
```

where:

- `W` = work
- `F` = force
- `d` = displacement
- `θ` = angle between force and displacement

Special Cases

θ = 0°

Force and displacement are in the same direction.

```
cos(0°) = 1
```

Therefore:

```
W = Fd
```

θ = 90°

Force and displacement are perpendicular.

```
cos(90°) = 0
```

Therefore:

```
W = 0
```

θ = 180°

Force and displacement are in opposite directions.

```
cos(180°) = -1
```

Therefore:

```
W = -Fd
```

Lesson 9: Example of Work at an Angle

A force of:

```
20 N
```

acts on an object at an angle of:

```
60°
```

to the direction of displacement.

The object moves:

```
5 m
```

Find the work done.

Given:

```
F = 20 N
d = 5 m
θ = 60°
```

Use:

```
W = Fd cos(θ)
```

Since:

```
cos(60°) = 0.5
```

we get:

```
W = 20 × 5 × 0.5
```

```
W = 50 J
```

Therefore, the work done is:

```
50 J
```

Lesson 10: Work Done by Gravity

Gravity can do work when an object changes height.

When an object falls downward, gravity and displacement are in the same direction.

Therefore, gravity does positive work.

For an object of mass `m` falling through a vertical distance `h`:

```
W = mgh
```

where:

- `m` = mass
- `g` = acceleration due to gravity
- `h` = vertical displacement

Example

A:

```
2 kg
```

object falls through:

```
5 m
```

Take:

```
g = 9.8 m/s²
```

Then:

```
W = mgh
```

```
W = 2 × 9.8 × 5
```

```
W = 98 J
```

Gravity does:

```
98 J
```

of positive work.

Lesson 11: Work Done Against Gravity

When an object is lifted upward at constant speed, an external force must act upward against gravity.

The external force does positive work on the object.

Gravity does negative work.

If an object of mass `m` is lifted through height `h`, the work done against gravity is:

```
W = mgh
```

This work is stored as gravitational potential energy.

Example

A student lifts a:

```
5 kg
```

object through:

```
2 m
```

Take:

```
g = 9.8 m/s²
```

Work done against gravity:

```
W = mgh
```

```
W = 5 × 9.8 × 2
```

```
W = 98 J
```

Lesson 12: Work and Kinetic Energy

Work can change an object's kinetic energy.

The **work-energy theorem** states:

```
Net Work = Change in Kinetic Energy
```

or:

```
W_net = ΔKE
```

Since:

```
KE = 1/2 mv²
```

we can write:

```
W_net = KE_final - KE_initial
```

Example

An object initially has:

```
KE = 20 J
```

After a force acts on it, its kinetic energy becomes:

```
KE = 50 J
```

Therefore:

```
W_net = 50 - 20
```

```
W_net = 30 J
```

The net work done is:

```
30 J
```

Lesson 13: Kinetic Energy

Kinetic energy is the energy an object possesses because of its motion.

The formula is:

```
KE = 1/2 mv²
```

where:

- `m` = mass
- `v` = velocity

Example

A:

```
2 kg
```

object moves at:

```
5 m/s
```

Therefore:

```
KE = 1/2(2)(5²)
```

```
KE = 25 J
```

Lesson 14: Potential Energy

Potential energy is stored energy associated with an object's position or configuration.

For gravitational potential energy near Earth's surface:

```
PE = mgh
```

where:

- `m` = mass
- `g` = gravitational acceleration
- `h` = height

Example

A:

```
3 kg
```

object is placed:

```
4 m
```

above the ground.

Using:

```
g = 9.8 m/s²
```

we get:

```
PE = 3 × 9.8 × 4
```

```
PE = 117.6 J
```

Lesson 15: Work and Potential Energy

When an object is lifted, work is done against gravity.

That work is stored as gravitational potential energy.

Therefore:

```
Work done against gravity = Increase in gravitational potential energy
```

So:

```
W = ΔPE
```

For a height increase `h`:

```
W = mgh
```

Lesson 16: Work and Mechanical Energy

Mechanical energy is the sum of kinetic and potential energy.

```
Mechanical Energy = KE + PE
```

or:

```
E_mechanical = KE + PE
```

In an ideal system with no energy losses:

```
KE + PE = constant
```

For example, when an object falls:

- Potential energy decreases.
- Kinetic energy increases.
- Total mechanical energy remains constant if air resistance and other losses are ignored.

Lesson 17: Work Done by Friction

Friction usually acts opposite to an object's displacement.

Therefore, friction does negative work.

If friction has magnitude `f` and the object moves a distance `d`:

```
W_friction = -fd
```

Example

A friction force of:

```
8 N
```

acts on an object moving:

```
10 m
```

Therefore:

```
W_friction = -(8)(10)
```

```
W_friction = -80 J
```

The negative sign indicates that friction removes mechanical energy from the object's motion.

Lesson 18: Net Work

An object may have several forces acting on it.

The total work done by all forces is called **net work**.

```
W_net = W₁ + W₂ + W₃ + ...
```

According to the work-energy theorem:

```
W_net = ΔKE
```

Example

Suppose:

- Applied force does `100 J` of work.
- Friction does `-30 J` of work.

Then:

```
W_net = 100 + (-30)
```

```
W_net = 70 J
```

Therefore, the net work is:

```
70 J
```

Lesson 19: Work Done by Multiple Forces

Consider a box moving horizontally.

Forces may include:

- Applied force
- Friction
- Gravity
- Normal force

If the box moves horizontally:

- Applied force may do positive work.
- Friction may do negative work.
- Gravity does zero work.
- Normal force does zero work.

The net work is the sum of the work done by all forces.

Lesson 20: Work and Force-Displacement Graphs

When force varies with displacement, work can be determined from the area under a force-displacement graph.

For a constant force:

```
Work = Area under F-d graph
```

For a rectangular graph:

```
W = F × d
```

For a triangular graph:

```
W = 1/2 × base × height
```

This provides a useful graphical method for determining work.

Lesson 21: Variable Force

In many real situations, force is not constant.

For example:

- Stretching a spring.
- Pushing an object with changing force.
- Moving an object through a changing resistance.

For a variable force, the work is represented by the area under the force-displacement graph.

At a more advanced level:

```
W = ∫ F dx
```

This means that work is found by summing small amounts of work over the displacement.

Lesson 22: Work Done by a Spring

A spring exerts a restoring force when stretched or compressed.

For an ideal spring, Hooke's law gives:

```
F = -kx
```

where:

- `F` = spring force
- `k` = spring constant
- `x` = displacement from equilibrium

The negative sign indicates that the spring force acts opposite to the displacement.

The work required to stretch a spring from zero displacement to `x` is:

```
W = 1/2 kx²
```

This energy is stored as elastic potential energy.

Lesson 23: Efficiency and Useful Work

In real systems, not all input energy becomes useful output.

Some energy may be transferred to the surroundings as:

- Heat

- Sound
- Vibration
- Frictional losses

Efficiency compares useful output energy with input energy.

```
Efficiency = Useful Output Energy / Input Energy × 100%
```

Example

A machine receives:

```
500 J
```

of energy and produces:

```
400 J
```

of useful output.

Efficiency:

```
Efficiency = (400/500) × 100
```

```
Efficiency = 80%
```

Lesson 24: Real-World Applications of Work

The concept of work is used in many areas.

Transportation

Work is done when engines provide forces that move vehicles.

Construction

Machines perform work when lifting, moving, or cutting materials.

Sports

Athletes perform work when applying forces that move their bodies or equipment.

Machines

Machines transfer energy through work.

Examples include:

- Cranes
- Elevators
- Motors
- Pumps
- Conveyor belts

Everyday Activities

Examples include:

- Lifting a bag.
- Pushing a chair.
- Pulling a suitcase.
- Climbing stairs.

Solved Problems

Problem 1: Basic Work

A force of 30 N moves an object 4 m in the direction of the force.

Find the work done.

Solution

```
W = Fd
```

```
W = 30 × 4
```

```
W = 120 J
```

Answer: 120 J

Problem 2: Negative Work

A friction force of 15 N acts on an object moving 8 m.

Find the work done by friction.

Solution

Friction acts opposite to displacement.

Therefore:

```
W = -Fd
```

```
W = -(15)(8)
```

```
W = -120 J
```

Answer: -120 J

Problem 3: Work at an Angle

A force of 50 N acts at an angle of 60° to the displacement. The object moves 10 m.

Find the work.

Solution

```
W = Fd cos(θ)
```

```
W = 50 × 10 × cos(60°)
```

Since:

```
cos(60°) = 0.5
```

```
W = 50 × 10 × 0.5
```

```
W = 250 J
```

Answer: 250 J

Problem 4: Work Against Gravity

A 10 kg object is lifted through a height of 3 m.

Take:

```
g = 9.8 m/s²
```

Find the work done against gravity.

Solution

```
W = mgh
```

```
W = 10 × 9.8 × 3
```

```
W = 294 J
```

Answer: 294 J

Problem 5: Net Work

An applied force does 200 J of work on an object while friction does -50 J.

Find the net work.

Solution

```
W_net = W_applied + W_friction
```

```
W_net = 200 + (-50)
```

```
W_net = 150 J
```

Answer: 150 J

Problem 6: Work-Energy Theorem

An object's kinetic energy increases from 30 J to 90 J.

Find the net work done.

Solution

```
W_net = ΔKE
```

```
W_net = KE_final - KE_initial
```

```
W_net = 90 - 30
```

```
W_net = 60 J
```

Answer: 60 J

Problem 7: Kinetic Energy

A 4 kg object moves at 5 m/s.

Find its kinetic energy.

Solution

```
KE = 1/2mv²
```

```
KE = 1/2(4)(5²)
```

```
KE = 2 × 25
```

```
KE = 50 J
```

Answer: 50 J

Practice Exercises

Exercise 1

A force of 25 N moves an object through 6 m in the direction of the force.

Find the work done.

Exercise 2

A force of 40 N acts on an object through a displacement of 5 m at an angle of 60°.

Find the work done.

Exercise 3

A friction force of 12 N acts over a displacement of 10 m.

Find the work done by friction.

Exercise 4

A 5 kg object is lifted through a height of 4 m.

Take:

```
g = 9.8 m/s²
```

Find the work done against gravity.

Exercise 5

A 2 kg object moves at 10 m/s.

Find its kinetic energy.

Exercise 6

An object's kinetic energy changes from 100 J to 160 J.

Find the net work done.

Exercise 7

An applied force performs 300 J of work while friction performs -80 J.

Find the net work.

Exercise 8

A machine receives 1000 J of input energy and produces 750 J of useful output energy.

Calculate its efficiency.

Exercise 9

Explain why a person holding a heavy object stationary does no mechanical work on the object.

Exercise 10

A person carries a bag horizontally at constant height. Explain why the upward force exerted by the person on the bag does zero mechanical work on the bag, assuming the force is vertical and the displacement is horizontal.

Mini Project

Investigating Work in Everyday Activities

Students will investigate a simple everyday activity involving work.

Possible activities include:

- Lifting a backpack.
- Pushing a box.
- Pulling a suitcase.
- Climbing stairs.
- Moving a chair.

Students should:

1. Identify the force involved.

2. Estimate or measure the displacement.
3. Determine the direction of force.
4. Determine the direction of displacement.
5. Calculate the work done.
6. Identify whether the work is positive, negative, or approximately zero.
7. Explain where the transferred energy goes.

For a lifting activity, students can use:

```
W = mgh
```

For a force acting along the displacement, students can use:

```
W = Fd
```

Students should present their results in a table.

Final Quiz

1. What is work in physics?
2. What two main quantities are required for mechanical work?
3. What is the SI unit of work?
4. Define one joule.
5. When is work positive?
6. When is work negative?
7. When is work zero?
8. What happens when force and displacement are perpendicular?
9. State the formula for work done by a constant force.
10. What is the formula for work when force acts at an angle?
11. What is the significance of the angle between force and displacement?
12. What is the work done when the angle is 0°?
13. What is the work done when the angle is 90°?
14. What is the work done when the angle is 180°?
15. How does friction usually affect work?
16. What is the work-energy theorem?
17. Define kinetic energy.
18. State the formula for kinetic energy.
19. Define gravitational potential energy.
20. State the formula for gravitational potential energy.
21. How is work related to potential energy?
22. What is net work?
23. How can work be determined from a force-displacement graph?
24. What is variable force?
25. What is Hooke's law?
26. What is elastic potential energy?
27. What is efficiency?
28. Why is the work done by friction usually negative?

29. Give two examples of positive work.
30. Give two examples of zero work.
31. Give one example of negative work.
32. Explain the relationship between work and energy.
33. What happens to potential energy when an object falls?
34. What happens to kinetic energy when a falling object speeds up?
35. Give three real-world applications of work.

Course Summary

After completing this course, students should understand that work in physics represents energy transfer caused by forces acting through displacement.

The major concepts covered are:

- Definition of work
- Force and displacement
- SI unit of work
- Joule
- Positive work
- Negative work
- Zero work
- Work at an angle
- Work done by gravity
- Work against gravity
- Kinetic energy
- Potential energy
- Work-energy theorem
- Net work
- Work done by friction
- Force-displacement graphs
- Variable force
- Work done by springs
- Elastic potential energy
- Efficiency
- Real-world applications

Key Formulas

```
W = Fd
```

```
W = Fd cos(θ)
```

```
W = mgh
```

```
KE = 1/2mv²
```

```
PE = mgh
```

```
W_net = ΔKE
```

```
W_spring = 1/2kx²
```

```
Efficiency = (Useful Output Energy / Input Energy) × 100%
```

Next Step

After completing **Physics – Work**, students can continue with **Energy and Power**, followed by **Force and Laws of Motion**, **Gravitation**, **Waves**, and **Electricity**.